

Qatar-1b

R-0350

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180928 · created 2026-07-18T09:48:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458389.7021 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.024
Transit depth (Rp/Rs)^2	1.66 +/- 0.61 [%]
Orbital Inclination (inc)	83.52 +/- 1.61
Airmass coefficient 1 (a1)	0.933 +/- 0.012
Airmass coefficient 2 (a2)	0.057 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [547, 415]
Optimal Aperture	2.78
Optimal Annulus	5.75
Transit Duration (day)	0.062 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.024 vs 0.1463 (deviation 0.72σ)
Duration fit vs published	0.062 d vs 0.0692 d (deviation 0.55σ)
Mid-time O–C	2.7 min
Depth SNR	5.4
Residual scatter	1.33 %

Light curve

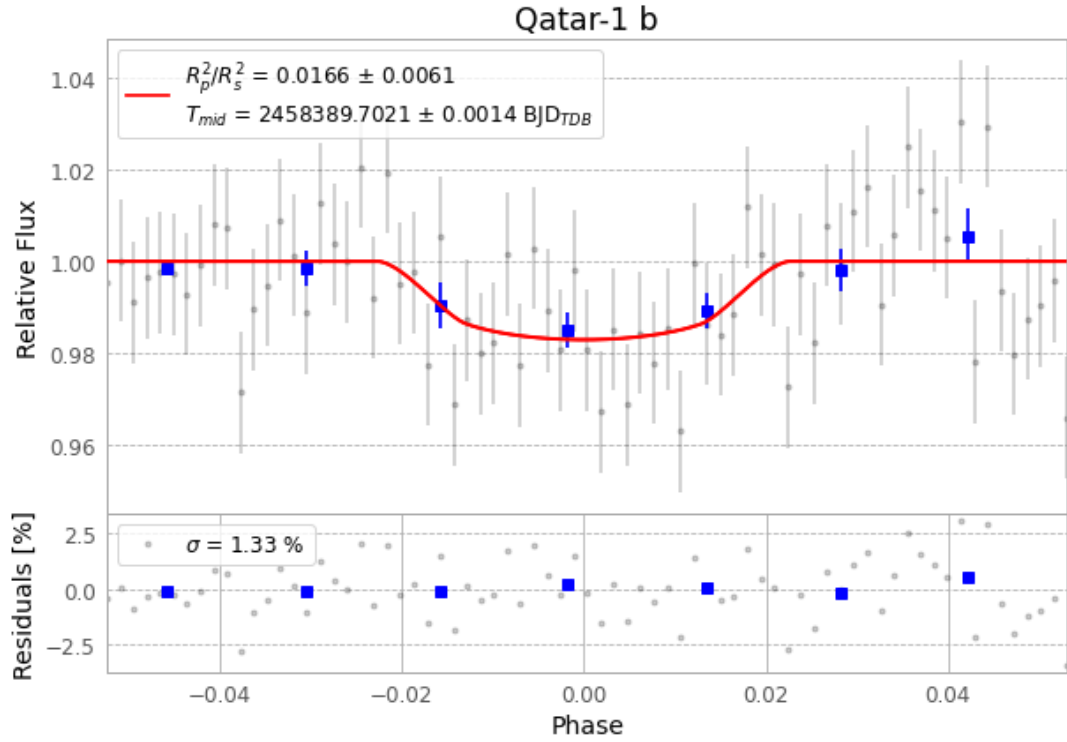


Figure 1 — FinalLightCurve_Qatar-1 b_2018-09-27.png (SHA-256 2673a63d20775af9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 336], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0bd26b7dfab1a370...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

73 FITS frames (48 MB), Qatar-1180928030012.FITS → Qatar-1180928063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180928.log (691b58767814...), FinalLightCurve_Qatar-1 b_2018-09-27.png (2673a63d2077...), FinalLightCurve_Qatar-1 b_2018-09-27.csv (078f55135630...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 09:48Z.